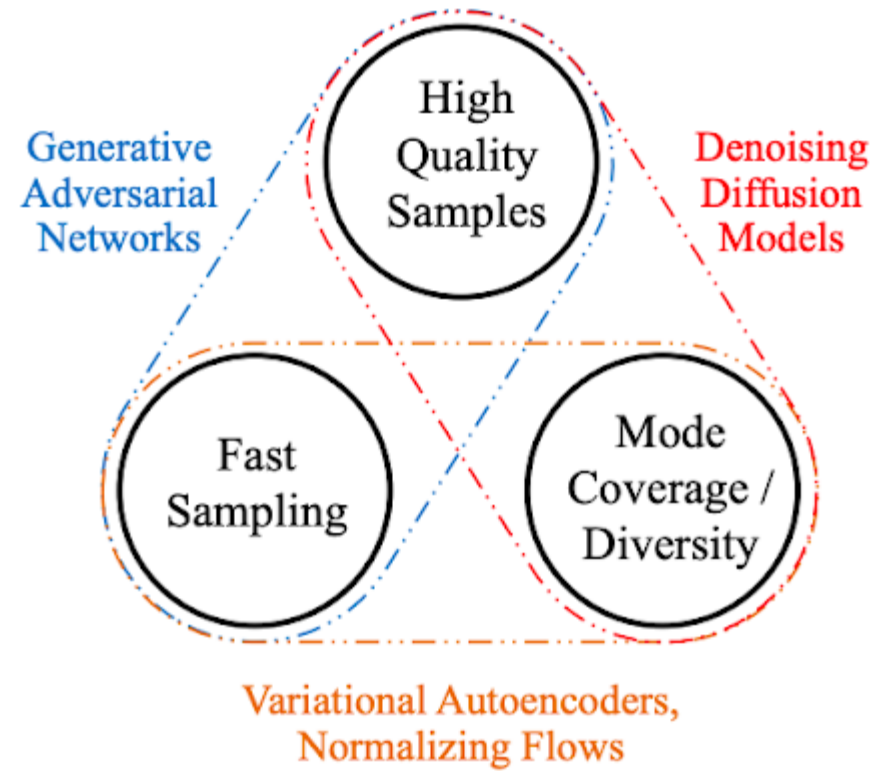


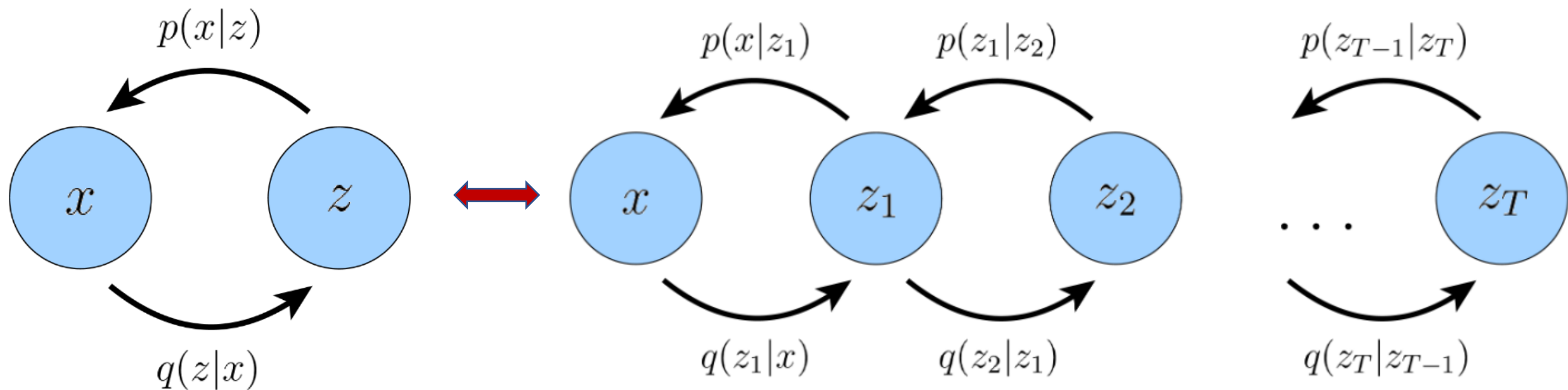
diffusion models

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10.1 diffusion models



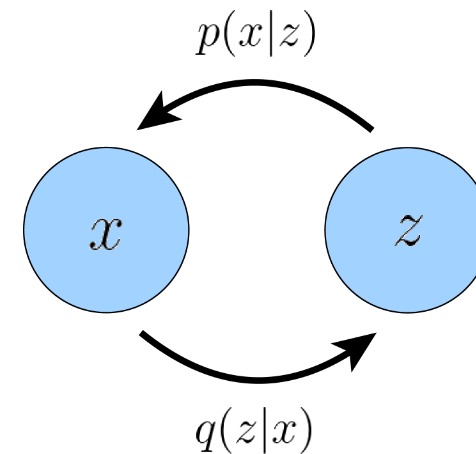
from VAE to hierarchical VAE



an excellent review by Google Brain

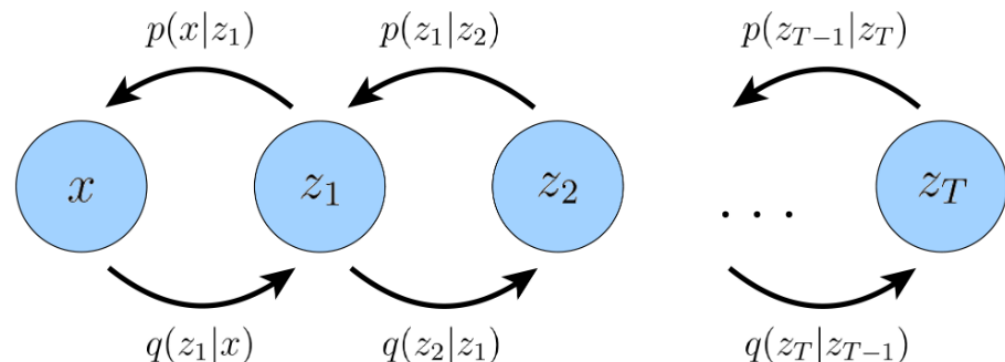
Luo, C. (2022). Understanding diffusion models: A unified perspective. *arXiv preprint arXiv:2208.11970*.

from VAE to hierarchical VAE



$$\begin{aligned} \text{ELBO} \quad \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p(x, z)}{q_{\phi}(z|x)} \right] &= \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x|z)p(z)}{q_{\phi}(z|x)} \right] \\ &= \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] + \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p(z)}{q_{\phi}(z|x)} \right] \\ &= \underbrace{\mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q_{\phi}(z|x) \parallel p(z))}_{\text{prior matching term}} \end{aligned}$$

from VAE to hierarchical VAE

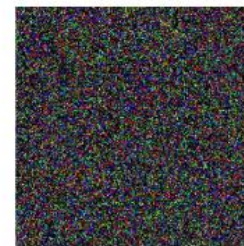
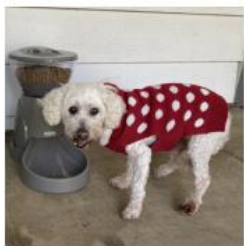
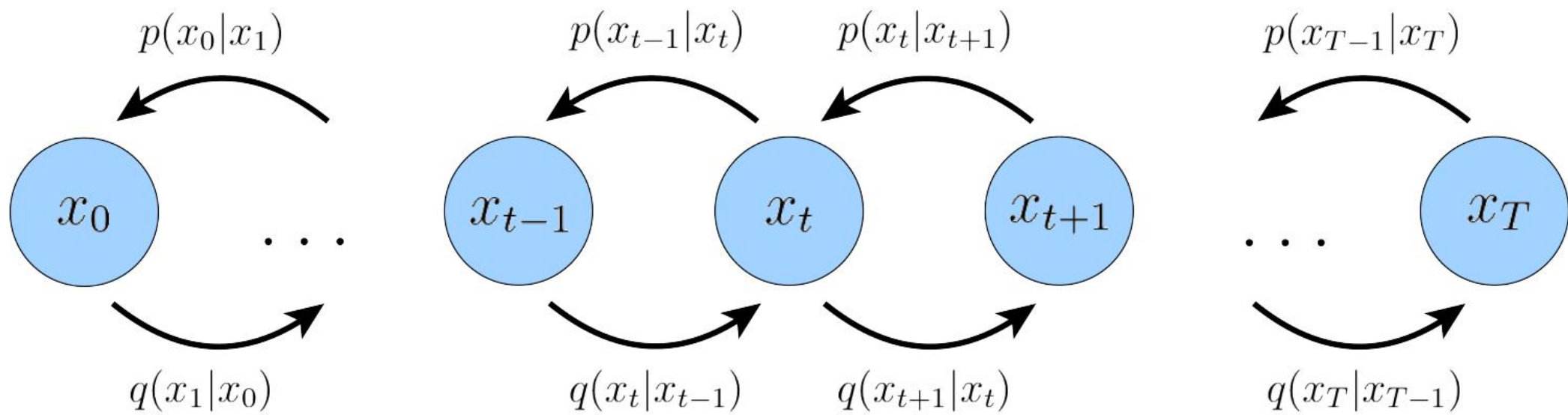


$$\begin{aligned}\log p(\mathbf{x}) &= \log \int p(\mathbf{x}, \mathbf{z}_{1:T}) d\mathbf{z}_{1:T} \\ &= \log \int \frac{p(\mathbf{x}, \mathbf{z}_{1:T}) q_\phi(\mathbf{z}_{1:T}|\mathbf{x})}{q_\phi(\mathbf{z}_{1:T}|\mathbf{x})} d\mathbf{z}_{1:T} \\ &= \log \mathbb{E}_{q_\phi(\mathbf{z}_{1:T}|\mathbf{x})} \left[\frac{p(\mathbf{x}, \mathbf{z}_{1:T})}{q_\phi(\mathbf{z}_{1:T}|\mathbf{x})} \right] \\ &\geq \mathbb{E}_{q_\phi(\mathbf{z}_{1:T}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z}_{1:T})}{q_\phi(\mathbf{z}_{1:T}|\mathbf{x})} \right]\end{aligned}$$

ELBO for hierarchical

$$= \mathbb{E}_{q_\phi(\mathbf{z}_{1:T}|\mathbf{x})} \left[\log \frac{p(z_T) p_\theta(\mathbf{x}|z_1) \prod_{t=2}^T p_\theta(z_{t-1}|z_t)}{q_\phi(z_1|\mathbf{x}) \prod_{t=2}^T q_\phi(z_t|z_{t-1})} \right]$$

diffusion models



diffusion models

- a **diffusion model** can be thought of as a hierarchical VAE with the following specified features:
 1. the **latent dimension** is **equal** to the **data dimension**
 2. the structure of the latent **encoder** at each timestep is not learned; it is **pre-defined as a linear Gaussian model** (i.e., in a diffusion model, at each timestep, inject a Gaussian noise)
 3. the distribution of the latent at final timestep T is a standard Gaussian $p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T | \mathbf{0}, I)$

forward diffusion process

- given $\mathbf{x}_0 \sim q(\mathbf{x})$, we define the encoder transition:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t | \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$$

- here, $\{\alpha_t\}_{t=1}^T$ is called the **variance schedule**
- note that $\beta_t = 1 - \alpha_t$ is simply the variance
- together,

$$q(\mathbf{x}_{1:T} | \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1})$$

forward diffusion process

- by reparameterization, $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t | \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1 - \alpha_t)\mathbf{I})$ can be put as

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1 - \alpha_t}\boldsymbol{\epsilon}_{t-1}$$

where $\boldsymbol{\epsilon}_{t-1}$ is a standard Gaussian noise $\boldsymbol{\epsilon}_{t-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

- this is a recursive relation which can be continued

forward diffusion process

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1}$$

$$= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}} \boldsymbol{\epsilon}_{t-2}) + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1}$$

$$= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \underbrace{\sqrt{\alpha_t - \alpha_t \alpha_{t-1}} \boldsymbol{\epsilon}_{t-2}} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1}$$



forward diffusion process

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1}$$

$$= \dots$$

$$= \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_0$$

where

$$\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

forward diffusion process

- now that $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}_0$
- we have $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t|\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$

backward diffusion process

- at the end of the forward diffusion steps, $p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T | \mathbf{0}, \mathbf{I})$
- of course, if we can derive $p(\mathbf{x}_{t-1} | \mathbf{x}_t)$, we will be like 🐱
- however, that would not be a simple task and we have to model those conditional densities using, (un)fortunately, e.g., neural networks

$$p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t)) \quad \text{difficult}$$

backward diffusion process

- on the other hand, when conditioned on the input image, the density is tractable with

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0) q(\mathbf{x}_{t-1} | \mathbf{x}_0)}{q(\mathbf{x}_t | \mathbf{x}_0)}$$



$$\propto \mathcal{N}(\mathbf{x}_{t-1}; \underbrace{\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\mathbf{x}_0}{1 - \bar{\alpha}_t}}_{\mu_q(\mathbf{x}_t, \mathbf{x}_0)}, \underbrace{\frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{I}}_{\Sigma_q(t) = \sigma_q^2(t) \mathbf{I}})$$

backward diffusion process

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$$

$$\propto \mathcal{N}(\mathbf{x}_{t-1}; \underbrace{\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\mathbf{x}_0}{1 - \bar{\alpha}_t}}_{\mu_q(\mathbf{x}_t, \mathbf{x}_0)}, \underbrace{\frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{I}}_{\Sigma_q(t)})$$

set to be the same

$$p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boxed{\boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t)})$$

loss function

- let's first look at the ELBO, whose maximization is the loss function

$$\begin{aligned}\log p(\mathbf{x}) &\geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] \\&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) \prod_{t=1}^T p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})} \right] \\&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) \prod_{t=2}^T p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_1|\mathbf{x}_0) \prod_{t=2}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})} \right] \\&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) \prod_{t=2}^T p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_1|\mathbf{x}_0) \prod_{t=2}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)} \right] \\&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_T) p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \prod_{t=2}^T \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)} \right] \\&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \prod_{t=2}^T \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}} \right]\end{aligned}$$

loss function

$$\begin{aligned}
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \prod_{t=2}^T \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}} \right] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)}{\cancel{q(\mathbf{x}_1|\mathbf{x}_0)}} + \log \frac{\cancel{q(\mathbf{x}_1|\mathbf{x}_0)}}{q(\mathbf{x}_T|\mathbf{x}_0)} + \log \prod_{t=2}^T \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_T|\mathbf{x}_0)} + \sum_{t=2}^T \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)] + \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} \right] + \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] \\
&= \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)] + \mathbb{E}_{q(\mathbf{x}_T|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} \right] + \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t, \mathbf{x}_{t-1}|\mathbf{x}_0)} \left[\log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] \\
&= \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{\text{prior matching term}} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\text{denoising matching term}}
\end{aligned}$$

loss function

- moreover,

$$\begin{aligned} & \arg \min_{\boldsymbol{\theta}} D_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \parallel p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1} | \mathbf{x}_t)) \\ &= \arg \min_{\boldsymbol{\theta}} D_{\text{KL}}(\mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_q, \boldsymbol{\Sigma}_q(t)) \parallel \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\boldsymbol{\theta}}, \boldsymbol{\Sigma}_q(t))) \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2} \left[\log \frac{|\boldsymbol{\Sigma}_q(t)|}{|\boldsymbol{\Sigma}_q(t)|} - d + \text{tr}(\boldsymbol{\Sigma}_q(t)^{-1} \boldsymbol{\Sigma}_q(t)) + (\boldsymbol{\mu}_{\boldsymbol{\theta}} - \boldsymbol{\mu}_q)^T \boldsymbol{\Sigma}_q(t)^{-1} (\boldsymbol{\mu}_{\boldsymbol{\theta}} - \boldsymbol{\mu}_q) \right] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2} [\log 1 - d + d + (\boldsymbol{\mu}_{\boldsymbol{\theta}} - \boldsymbol{\mu}_q)^T \boldsymbol{\Sigma}_q(t)^{-1} (\boldsymbol{\mu}_{\boldsymbol{\theta}} - \boldsymbol{\mu}_q)] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2} [(\boldsymbol{\mu}_{\boldsymbol{\theta}} - \boldsymbol{\mu}_q)^T \boldsymbol{\Sigma}_q(t)^{-1} (\boldsymbol{\mu}_{\boldsymbol{\theta}} - \boldsymbol{\mu}_q)] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2} [(\boldsymbol{\mu}_{\boldsymbol{\theta}} - \boldsymbol{\mu}_q)^T (\sigma_q^2(t) \mathbf{I})^{-1} (\boldsymbol{\mu}_{\boldsymbol{\theta}} - \boldsymbol{\mu}_q)] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} [\|\boldsymbol{\mu}_{\boldsymbol{\theta}} - \boldsymbol{\mu}_q\|_2^2] \end{aligned}$$

loss function

$$\begin{aligned} & \arg \min_{\boldsymbol{\theta}} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)) \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \left[\|\boldsymbol{\mu}_{\boldsymbol{\theta}} - \boldsymbol{\mu}_q\|_2^2 \right] \end{aligned}$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \propto \mathcal{N}(\mathbf{x}_{t-1}; \underbrace{\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\mathbf{x}_0}{1 - \bar{\alpha}_t}}_{\boldsymbol{\mu}_q(\mathbf{x}_t, \mathbf{x}_0)}, \underbrace{\frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{I}}_{\boldsymbol{\Sigma}_q(t) = \sigma_q^2(t)\mathbf{I}})$$

loss function

- we can model $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t))$ so that

$$\boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t)}{1 - \bar{\alpha}_t}$$

loss function

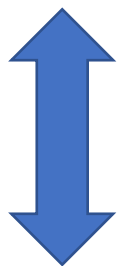
- the optimization for the KL divergence at time t thus reduces to

$$\begin{aligned} & \arg \min_{\boldsymbol{\theta}} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)) \\ &= \arg \min_{\boldsymbol{\theta}} D_{\text{KL}}(\mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_q, \boldsymbol{\Sigma}_q(t)) \parallel \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\boldsymbol{\theta}}, \boldsymbol{\Sigma}_q(t))) \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \left[\left\| \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\hat{\mathbf{x}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)}{1 - \bar{\alpha}_t} - \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\mathbf{x}_0}{1 - \bar{\alpha}_t} \right\|_2^2 \right] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \left[\left\| \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\hat{\mathbf{x}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)}{1 - \bar{\alpha}_t} - \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\mathbf{x}_0}{1 - \bar{\alpha}_t} \right\|_2^2 \right] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \left[\left\| \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} (\hat{\mathbf{x}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \mathbf{x}_0) \right\|_2^2 \right] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \frac{\bar{\alpha}_{t-1}(1 - \alpha_t)^2}{(1 - \bar{\alpha}_t)^2} \left[\|\hat{\mathbf{x}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \mathbf{x}_0\|_2^2 \right] \end{aligned}$$

loss function

- to summarize, the loss in diffusion models reduces to reconstructions

$$\underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{\text{prior matching term}} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\text{denoising matching term}}$$



reconstruction loss
same as other t



nothing to learn



$$\frac{1}{2\sigma_q^2(t)} \frac{\bar{\alpha}_{t-1}(1 - \alpha_t)^2}{(1 - \bar{\alpha}_t)^2} \left[\|\hat{\mathbf{x}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \mathbf{x}_0\|_2^2 \right]$$

loss function: a different choice

- rather than reconstructing the input, we can reconstruct the noises
- to see why, from $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}_0$,



$$\mathbf{x}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}_0}{\sqrt{\bar{\alpha}_t}}$$

loss function: a different choice

$$\begin{aligned}
 \mu_q(\mathbf{x}_t, \mathbf{x}_0) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\mathbf{x}_0}{1 - \bar{\alpha}_t} \\
 &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t) \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}_0}{\sqrt{\bar{\alpha}_t}}}{1 - \bar{\alpha}_t} \\
 &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + (1 - \alpha_t) \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}_0}{\sqrt{\alpha_t}}}{1 - \bar{\alpha}_t} \\
 &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t}{1 - \bar{\alpha}_t} + \frac{(1 - \alpha_t)\mathbf{x}_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} - \frac{(1 - \alpha_t)\sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}_0}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \\
 &= \left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} + \frac{1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \right) \mathbf{x}_t - \frac{(1 - \alpha_t)\sqrt{1 - \bar{\alpha}_t}}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \boldsymbol{\epsilon}_0 \\
 &= \left(\frac{\alpha_t(1 - \bar{\alpha}_{t-1})}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} + \frac{1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \right) \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}} \boldsymbol{\epsilon}_0 \\
 &= \frac{\alpha_t - \bar{\alpha}_t + 1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}} \boldsymbol{\epsilon}_0 \\
 &= \frac{1 - \bar{\alpha}_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}} \boldsymbol{\epsilon}_0 \\
 &= \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}} \boldsymbol{\epsilon}_0
 \end{aligned}$$

now in terms of $\mathbf{x}_t$ and $\boldsymbol{\epsilon}_0$, instead of $\mathbf{x}_t$ and $\mathbf{x}_0$!

loss function: a different choice

- instead of setting

$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t)}{1 - \bar{\alpha}_t}$$

- we can choose to set

$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}}\hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$

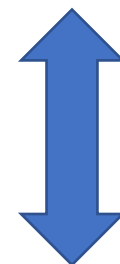
loss function: a different choice

- correspondingly,

$$\begin{aligned} & \arg \min_{\boldsymbol{\theta}} D_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \parallel p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1} | \mathbf{x}_t)) \\ &= \arg \min_{\boldsymbol{\theta}} D_{\text{KL}}(\mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_q, \boldsymbol{\Sigma}_q(t)) \parallel \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\boldsymbol{\theta}}, \boldsymbol{\Sigma}_q(t))) \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \left[\left\| \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \boldsymbol{\epsilon}_0 \right\|_2^2 \right] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \left[\left\| \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \boldsymbol{\epsilon}_0 - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|_2^2 \right] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \left[\left\| \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} (\boldsymbol{\epsilon}_0 - \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)) \right\|_2^2 \right] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \frac{(1 - \alpha_t)^2}{(1 - \bar{\alpha}_t) \alpha_t} \left[\left\| \boldsymbol{\epsilon}_0 - \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|_2^2 \right] \end{aligned}$$

loss function: a different choice

$$\underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{\text{prior matching term}} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\text{denoising matching term}}$$



$$\frac{1}{2\sigma_q^2(t)} \frac{(1 - \alpha_t)^2}{(1 - \bar{\alpha}_t)\alpha_t} \left[\|\boldsymbol{\epsilon}_0 - \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\|_2^2 \right]$$

loss function: a different choice

$$\underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{\text{prior matching term}} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\text{denoising matching term}}$$



$$\underbrace{\frac{1}{2\sigma_q^2(t)} \frac{(1 - \alpha_t)^2}{(1 - \bar{\alpha}_t)\alpha_t}}_{\text{weight}} \left[\|\boldsymbol{\epsilon}_0 - \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\|_2^2 \right]$$

a simplified version ignores this weight when sampling t

algorithm for fitting the noise: DDPM

Algorithm 1 Training

```
1: repeat  
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$   
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$   
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
5:   Take gradient descent step on  
        $\nabla_{\theta} \|\epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, t)\|^2$   
6: until converged
```

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
2: for  $t = T, \dots, 1$  do  
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$   
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$   
5: end for  
6: return  $\mathbf{x}_0$ 
```

$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$

results of the DDPM

Table 1: CIFAR10 results. NLL measured in bits/dim.

Model	IS	FID	NLL Test (Train)
Conditional			
EBM [11]	8.30	37.9	
JEM [17]	8.76	38.4	
BigGAN [3]	9.22	14.73	
StyleGAN2 + ADA (v1) [29]	10.06	2.67	
Unconditional			
Diffusion (original) [53]			≤ 5.40
Gated PixelCNN [59]	4.60	65.93	3.03 (2.90)
Sparse Transformer [7]			2.80
PixelIQN [43]	5.29	49.46	
EBM [11]	6.78	38.2	
NCSNv2 [56]		31.75	
NCSN [55]	8.87 ± 0.12	25.32	
SNGAN [39]	8.22 ± 0.05	21.7	
SNGAN-DDLS [4]	9.09 ± 0.10	15.42	
StyleGAN2 + ADA (v1) [29]	9.74 ± 0.05	3.26	
Ours (L , fixed isotropic Σ)	7.67 ± 0.13	13.51	≤ 3.70 (3.69)
Ours (L_{simple})	9.46 ± 0.11	3.17	≤ 3.75 (3.72)

Table 2: Unconditional CIFAR10 reverse process parameterization and training objective ablation. Blank entries were unstable to train and generated poor samples with out-of-range scores.

Objective	IS	FID
$\tilde{\mu}$ prediction (baseline)		
L , learned diagonal Σ	7.28 ± 0.10	23.69
L , fixed isotropic Σ	8.06 ± 0.09	13.22
$\ \tilde{\mu} - \tilde{\mu}_{\theta}\ ^2$	–	–
ϵ prediction (ours)		
L , learned diagonal Σ	–	–
L , fixed isotropic Σ	7.67 ± 0.13	13.51
$\ \tilde{\epsilon} - \epsilon_{\theta}\ ^2$ (L_{simple})	9.46 ± 0.11	3.17

Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33, 6840–6851.

DDPM sampling can be slow

- recall that $\mathbf{x}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_0}{\sqrt{\bar{\alpha}_t}}$
- if we use $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ in place of $\boldsymbol{\epsilon}_0$,
 - then $\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}}$
 - we can express $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
```

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DDIM

DENOISING DIFFUSION IMPLICIT MODELS

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- <https://colab.research.google.com/drive/12zJRGsh529tX2I8KVWMKn7WFjyP1wiYY?usp=sharing>

10.2 score-based diffusion

loss function: a third choice

- Tweedie's formula:
 - if $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then

$$\mathbb{E}[\boldsymbol{\mu}|\mathbf{z}] = \mathbf{z} + \boldsymbol{\Sigma} \nabla \log p(\mathbf{z})$$

loss function: a third choice

- recall

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

- applying Tweedie's formula yields

$$\sqrt{\bar{\alpha}_t}\mathbf{x}_0 = \mathbf{x}_t + (1 - \bar{\alpha}_t)\nabla \log p(\mathbf{x}_t)$$

$$\mathbf{x}_0 = \frac{\mathbf{x}_t + (1 - \bar{\alpha}_t)\nabla \log p(\mathbf{x}_t)}{\sqrt{\bar{\alpha}_t}}$$

loss function: a third choice

$$\begin{aligned}\mu_q(\mathbf{x}_t, \mathbf{x}_0) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\mathbf{x}_0}{1 - \bar{\alpha}_t} \\&= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t) \frac{\mathbf{x}_t + (1 - \bar{\alpha}_t) \nabla \log p(\mathbf{x}_t)}{\sqrt{\bar{\alpha}_t}}}{1 - \bar{\alpha}_t} \\&= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + (1 - \alpha_t) \frac{\mathbf{x}_t + (1 - \bar{\alpha}_t) \nabla \log p(\mathbf{x}_t)}{\sqrt{\alpha_t}}}{1 - \bar{\alpha}_t} \\&= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t}{1 - \bar{\alpha}_t} + \frac{(1 - \alpha_t)\mathbf{x}_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} + \frac{(1 - \alpha_t)(1 - \bar{\alpha}_t) \nabla \log p(\mathbf{x}_t)}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \\&= \left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} + \frac{1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \right) \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \nabla \log p(\mathbf{x}_t) \\&= \left(\frac{\alpha_t(1 - \bar{\alpha}_{t-1})}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} + \frac{1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \right) \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \nabla \log p(\mathbf{x}_t) \\&= \frac{\alpha_t - \bar{\alpha}_t + 1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \nabla \log p(\mathbf{x}_t) \\&= \frac{1 - \bar{\alpha}_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \nabla \log p(\mathbf{x}_t) \\&= \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \nabla \log p(\mathbf{x}_t)\end{aligned}$$

loss function: a third choice

$$\begin{aligned}
 \mu_q(\mathbf{x}_t, \mathbf{x}_0) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\mathbf{x}_0}{1 - \bar{\alpha}_t} \\
 &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t) \frac{\mathbf{x}_t + (1 - \bar{\alpha}_t) \nabla \log p(\mathbf{x}_t)}{\sqrt{\bar{\alpha}_t}}}{1 - \bar{\alpha}_t} \\
 &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + (1 - \alpha_t) \frac{\mathbf{x}_t + (1 - \bar{\alpha}_t) \nabla \log p(\mathbf{x}_t)}{\sqrt{\alpha_t}}}{1 - \bar{\alpha}_t} \\
 &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t}{1 - \bar{\alpha}_t} + \frac{(1 - \alpha_t)\mathbf{x}_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} + \frac{(1 - \alpha_t)(1 - \bar{\alpha}_t) \nabla \log p(\mathbf{x}_t)}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \\
 &= \left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} + \frac{1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \right) \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \nabla \log p(\mathbf{x}_t) \\
 &= \left(\frac{\alpha_t(1 - \bar{\alpha}_{t-1})}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} + \frac{1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \right) \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \nabla \log p(\mathbf{x}_t) \\
 &= \frac{\alpha_t - \bar{\alpha}_t + 1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \nabla \log p(\mathbf{x}_t) \\
 &= \frac{1 - \bar{\alpha}_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \nabla \log p(\mathbf{x}_t) \\
 &= \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \boxed{\nabla \log p(\mathbf{x}_t)}_{\text{score}}
 \end{aligned}$$

our third choice is to assume

$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} s_{\theta}(\mathbf{x}_t, t)$$

loss function: a third choice

$$\begin{aligned} & \arg \min_{\boldsymbol{\theta}} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)) \\ &= \arg \min_{\boldsymbol{\theta}} D_{\text{KL}}(\mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_q, \boldsymbol{\Sigma}_q(t)) \parallel \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\boldsymbol{\theta}}, \boldsymbol{\Sigma}_q(t))) \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \left[\left\| \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}} \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{\alpha_t}} \nabla \log p(\mathbf{x}_t) \right\|_2^2 \right] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \left[\left\| \frac{1-\alpha_t}{\sqrt{\alpha_t}} \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \frac{1-\alpha_t}{\sqrt{\alpha_t}} \nabla \log p(\mathbf{x}_t) \right\|_2^2 \right] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \left[\left\| \frac{1-\alpha_t}{\sqrt{\alpha_t}} (\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \nabla \log p(\mathbf{x}_t)) \right\|_2^2 \right] \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \frac{(1-\alpha_t)^2}{\alpha_t} \left[\left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \nabla \log p(\mathbf{x}_t) \right\|_2^2 \right] \end{aligned}$$

score-based generation

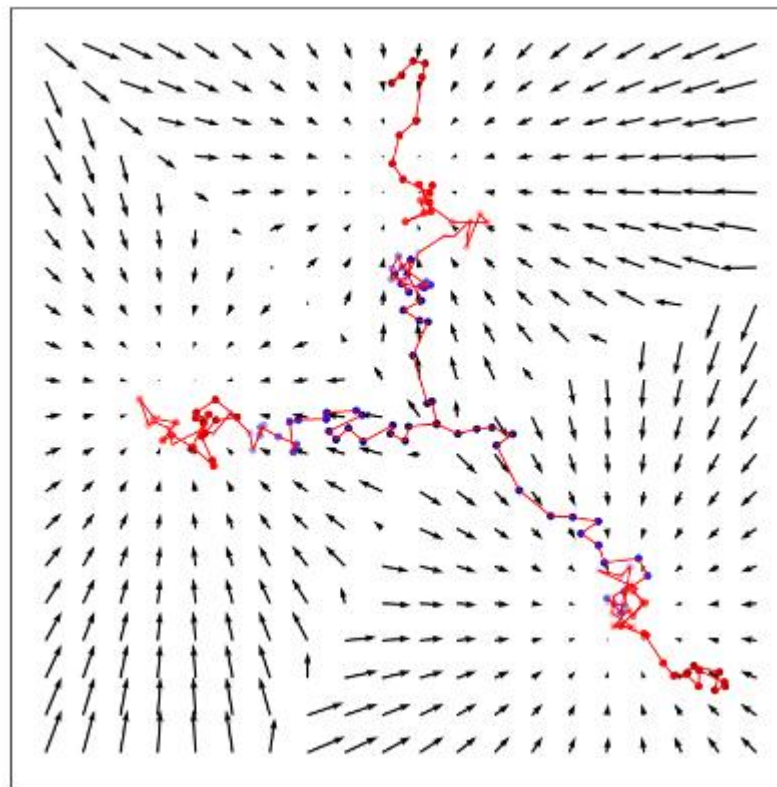
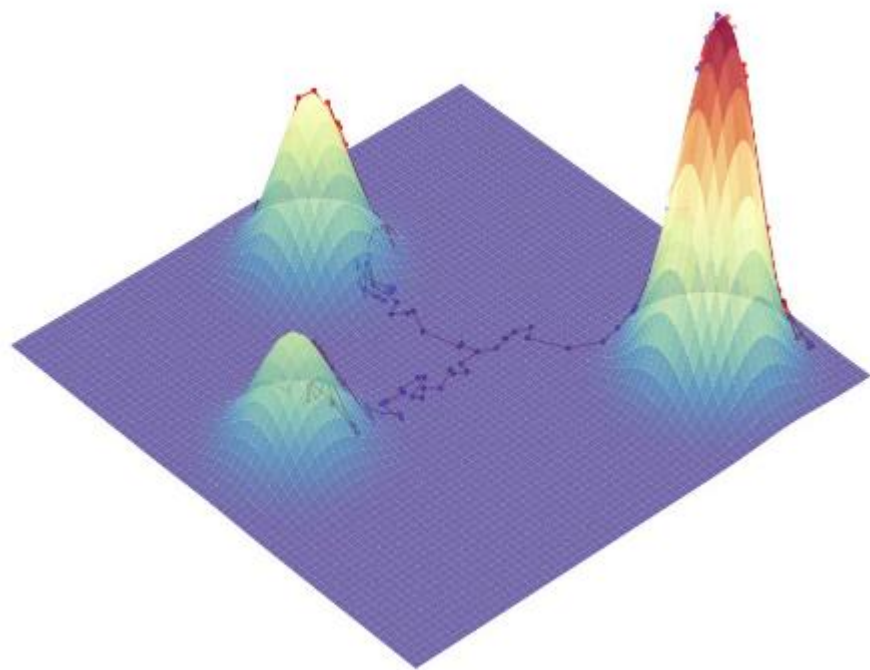


image credit: Luo, C. (2022). Understanding diffusion models: A unified perspective.
arXiv preprint arXiv:2208.11970.

what is the score? a revisit of derivation

$$\begin{aligned}
\mu_q(\mathbf{x}_t, \mathbf{x}_0) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\mathbf{x}_0}{1 - \bar{\alpha}_t} \\
&= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_0}{\sqrt{\bar{\alpha}_t}}}{1 - \bar{\alpha}_t} \\
&= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + (1 - \alpha_t)\frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_0}{\sqrt{\bar{\alpha}_t}}}{1 - \bar{\alpha}_t} \\
&= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t}{1 - \bar{\alpha}_t} + \frac{(1 - \alpha_t)\mathbf{x}_t}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} - \frac{(1 - \alpha_t)\sqrt{1 - \bar{\alpha}_t}\epsilon_0}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} \\
&= \left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} + \frac{1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} \right) \mathbf{x}_t - \frac{(1 - \alpha_t)\sqrt{1 - \bar{\alpha}_t}}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} \epsilon_0 \\
&= \left(\frac{\alpha_t(1 - \bar{\alpha}_{t-1})}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} + \frac{1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} \right) \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\bar{\alpha}_t}} \epsilon_0 \\
&= \frac{\alpha_t - \bar{\alpha}_t + 1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\bar{\alpha}_t}} \epsilon_0 \\
&= \frac{1 - \bar{\alpha}_t}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\bar{\alpha}_t}} \epsilon_0 \\
&= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\bar{\alpha}_t}} \epsilon_0
\end{aligned}$$

$$\begin{aligned}
\mu_q(\mathbf{x}_t, \mathbf{x}_0) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\mathbf{x}_0}{1 - \bar{\alpha}_t} \\
&= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)\frac{\mathbf{x}_t + (1 - \bar{\alpha}_t)\nabla \log p(\mathbf{x}_t)}{\sqrt{\bar{\alpha}_t}}}{1 - \bar{\alpha}_t} \\
&= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t + (1 - \alpha_t)\frac{\mathbf{x}_t + (1 - \bar{\alpha}_t)\nabla \log p(\mathbf{x}_t)}{\sqrt{\bar{\alpha}_t}}}{1 - \bar{\alpha}_t} \\
&= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})\mathbf{x}_t}{1 - \bar{\alpha}_t} + \frac{(1 - \alpha_t)\mathbf{x}_t}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} + \frac{(1 - \alpha_t)(1 - \bar{\alpha}_t)\nabla \log p(\mathbf{x}_t)}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} \\
&= \left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} + \frac{1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} \right) \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\bar{\alpha}_t}} \nabla \log p(\mathbf{x}_t) \\
&= \left(\frac{\alpha_t(1 - \bar{\alpha}_{t-1})}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} + \frac{1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} \right) \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\bar{\alpha}_t}} \nabla \log p(\mathbf{x}_t) \\
&= \frac{\alpha_t - \bar{\alpha}_t + 1 - \alpha_t}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\bar{\alpha}_t}} \nabla \log p(\mathbf{x}_t) \\
&= \frac{1 - \bar{\alpha}_t}{(1 - \bar{\alpha}_t)\sqrt{\bar{\alpha}_t}} \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\bar{\alpha}_t}} \nabla \log p(\mathbf{x}_t) \\
&= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t + \frac{1 - \alpha_t}{\sqrt{\bar{\alpha}_t}} \nabla \log p(\mathbf{x}_t)
\end{aligned}$$

score-based generation not following DDPM

- Langevin sampling (as an MCMC method)

$$d\mathbf{x} = \frac{1}{2} \nabla \log p(\mathbf{x}) dt + d\mathbf{w}$$

- this can be discretized using Euler(-Maruyama) method:

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{1}{2} \nabla \log p(\mathbf{x}_t) 2\eta + \sqrt{2\eta} \boldsymbol{\epsilon}_t$$

score-based generation not following DDPM

- stochastic differential equation (SDE)

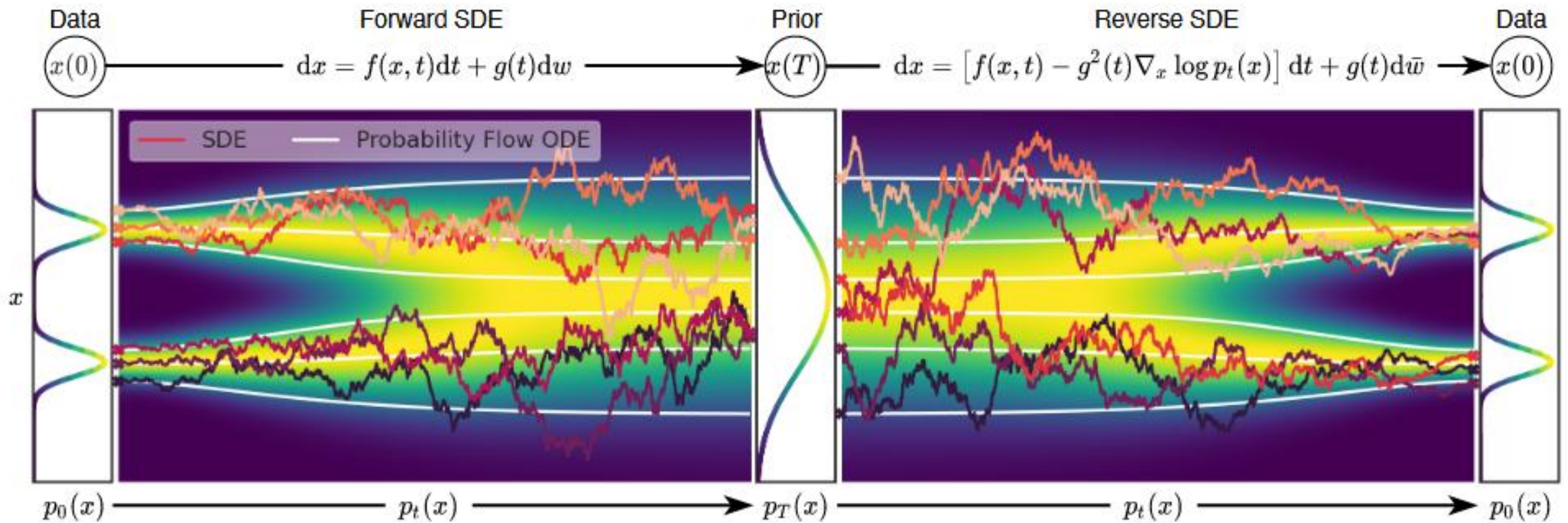
$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

- $\mathbf{f}: \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}^d$ is called the drift coefficient
- $g: \mathbb{R} \rightarrow \mathbb{R}$ is called the diffusion coefficient

- corresponding reverse SDE

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g(t)^2 \nabla \log p_t(\mathbf{x})]dt + g(t)d\mathbf{w}$$

score-based generation not following DDPM



score-based generation not following DDPM
but DDPM serves as a specific example

- $\mathbf{x}_i = \sqrt{\alpha_i}\mathbf{x}_{i-1} + \sqrt{1 - \alpha_i}\mathbf{z}_{i-1}$ is a discrete process that corresponds to

$$d\mathbf{x} = -\frac{1}{2}(1 - \alpha(t))\mathbf{x}dt + \sqrt{1 - \alpha(t)}d\mathbf{w}$$

Thank you!

Reference

- Luo, C. (2022). Understanding diffusion models: A unified perspective. *arXiv preprint arXiv:2208.11970*.
- Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33, 6840-6851.
- <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/#forward-diffusion-process>

